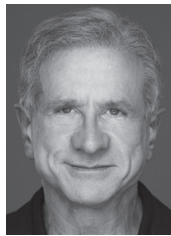

What executives need to know about knowledge management, large language models and generative AI

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Abstract This paper discusses the opportunities and risks presented by large language models (LLMs), which power the popular and widely adopted Chat-GPT types of applications. The potential benefits include support for enhancing the customer journey and efficient management of an ever-increasing volume of information for employees. Risks include hallucinations (made up answers by generative AI that are not factually correct), exposure of corporate intellectual property (IP) to training models, lack of traceability and audit trails and misalignment with brand guidelines. The approach to handling risk described in this paper is retrieval-augmented generation (RAG), which references corporate knowledge and data sources in order to identify precise answers and retrieve exactly what users want. The paper also outlines the need for a knowledge architecture which enables enriched embeddings into vector databases which retain the context of intelligently componentised content. Using RAG requires knowledge hygiene and metadata models, and the paper discusses an experiment in which results were measured with and without the knowledge architecture. The improvement was significant: 53 per cent of questions were answered correctly without the model versus 83 per cent with the model. The use of RAG virtually eliminated hallucinations, secured corporate IP and provided traceability and an audit trail.

KEYWORDS: RAG, retrieval augmented generation, generative AI, ChatGPT, LLMs, large language models, KM, knowledge management, LLM challenges, LLM solutions, knowledge models, metadata models, knowledge architecture

INTRODUCTION

The customer journey is a knowledge journey. Every time a customer interacts with an organisation, they seek answers to their questions: What products do you have? Which ones are best for me? How do I decide on a purchase? How/where can I purchase your product? How do I maintain it? How can it be serviced and supported? Customers need answers, and if they cannot find the answer on the company's website, they either call customer service or turn to a competitor.

Increasingly, organisations are offering better performing chatbots to deflect calls from the call centre, and many are now looking at generative artificial intelligence (AI) to power those bots, but both of those options have risks. Generative AI can support conversational search, product selection, support questions and more, and therefore can impact every stage of the journey. Generative AI can also support a more personalised experience. Figure 1 illustrates example questions that customers have along their journey.

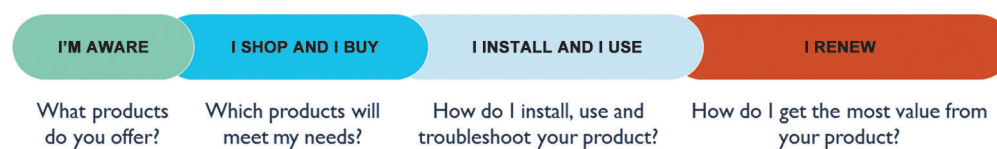
Applying generative AI is not as simple as it may seem, or as easy to implement as many vendors claim. The stampede of vendors entering the large language model (LLM)/ChatGPT market is likely to produce a deluge of failed projects, wasted

efforts and damaged careers because neither the vendors nor their customers understand the risks or how to counteract them to achieve success.

Executives and line of business managers are trying to factor in how LLMs and generative AI will be part of their current and future capabilities and figure out how transformation roadmaps will be impacted. The risks of diving into this technology are significant, and include:

- unrealistic expectations of LLMs as a magic solution to managing corporate content without human involvement;
- generating responses not aligned with company policies or brand;
- the knowledge gap that occurs when LLMs are not trained with current knowledge or knowledge of the organisation and therefore cannot produce answers — or worse, make up answers;
- difficulty distinguishing between creative outputs and fabricated responses (hallucinations);
- absence of clear audit trails and citation sources;
- decisions around training models: balancing usefulness with the threat of exposing trade secrets or other proprietary knowledge;

Every experience is a knowledge and content experience



EVERY QUESTION AND EVERY ANSWER NEEDS TO BE BROKEN DOWN INTO FUNDAMENTAL COMPONENTS.

Figure 1: The customer journey is a knowledge journey
Source: Earley Information Science, Inc., 2023

- potential financial burden of using proprietary LLMs and related enterprise software platforms.

A recent ZDNet article stated that ‘75% of businesses worldwide currently are implementing or considering plans to prohibit ChatGPT and other generative AI applications in their workplace. Of these, 61% said such measures will be permanent or long-term’. The article went on to cite risks associated with data security, privacy and brand reputation as reasons for the ban.¹ All of these are key issues in marketing and customer service and, more broadly, for the organisation as a whole.

But does avoiding these new technologies out of fear of risk engender a bigger risk — that of losing the potential benefits by ignoring LLMs completely? Getting LLMs right will increase productivity, save time on routine tasks, improve human creativity, act as a sounding board or starting point for research and help to synthesise and summarise large amounts of information. For many organisations, they can provide solid returns if they are used correctly.

WHAT ARE LLMS?

LLMs are complex constructs that contain word, concept and term relationships, knowledge of how language functions and feedback mechanisms that continually optimise outputs during the model training. They ultimately produce a neural network that can interpret questions from customers and predict the answers most likely to be responsive. LLMs are mathematical algorithms that manipulate words, sentences and documents in a mathematical representation called a vector.

Since it is difficult to think in more than three dimensions, imagine a vector in the x, y and z axes of a graph. By considering the dimension of time, it is possible to conceive of four dimensions. But vectors live in a multidimensional space, with each dimension

representing a characteristic of the content. For example, when comparing ‘man to king’, a vector representation would allow for the inference that comparing ‘woman to queen’ is analogous, since those vectors would be near one another. A syntactic relationship might compare vectors of ‘walk’ to ‘walking’ and ‘swim’ to . . . ? Based on the nearest vector the system would return ‘swimming’. The LLM contains word embeddings that represent the relationship of one term to another and the meaning of strings of words.

Other elements of the LLM enable its functionality. For example, positional encodings model the position of words in a sequence. Attention mechanisms allow the model to focus on different parts of the input text in order to compare it with other vectors and begin to optimise the output. LLMs use additional mechanisms to make sense of text and produce meaningful output. The number of parameters of a model can go into the hundreds of billions. This number includes the various weights, biases and embeddings — the higher the number of parameters, the greater language complexity the model can handle.²

THE VALUE OF LLMS

LLM-based applications can assist organisations in making sense of large volumes of documents and other content. However, using LLMs is not as simple as pointing them to an information source. Content needs to be correctly ingested to have maximal value. When this is done correctly, employees and customers will spend less time searching for what they need, and in many cases, the applications can anticipate a need and surface helpful content in the context of a task. Enabling personalisation will improve the customer experience and increase engagement as the result of getting people exactly what they need in the context in which they need it.

PROBLEMS WITH LLMS

It is important to understand the limitations of LLMS. They are not solutions in themselves but rather a part of a tool kit. Human intervention is needed at certain key points. In addition, the system needs to understand things that may not be in the public domain or policies that are specific to the organisation. That information has to be structured and curated. Humans need to capture and codify knowledge so it can be used in AI systems, and humans are also needed to solve new problems that an LLM cannot yet solve. LLMS do not automatically know your company's language, terminology, acronyms or processes.

A generalised LLM will attempt to answer questions where it does not have the information by creating (fabricating) an answer.³ This is because the model presents words and phrases that are statistically likely to go together but may not have a basis in fact. For example, using the LLM's understanding of language will likely not include organisation-specific processes or knowledge. Terminology may be unique to the enterprise. An LLM will not be able to make sense of information it does not possess.

RETRIEVAL-AUGMENTED GENERATION (RAG)

Because an LLM is trained on generally available information, competitive or proprietary intellectual property (IP) will not be part of the training content. Therefore, company data structures, knowledge and terminology (embodied in an ontology and, when associated with data, structured in a knowledge graph) needs to be referenced by the LLM. There is always information that needs to be protected from disclosure, including competitive strategies and knowledge of *customer needs and details* on precisely how the organisation serves those needs. That knowledge is the foundation for competitive differentiation, and the LLM needs access to it in order to provide

answers. At the same time, that information has to be safely processed by a commercial (at the application programming interface [API] level) or open source (in a private cloud or behind a firewall on-prem) LLM so that it does not become part of the LLM's training data.

Figure 2 compares the two approaches: the LLM providing the answers versus a knowledge base reference for 'ground truth'.

The solution is to use a localised/private cloud copy of an LLM or use APIs to access the commercial LLM and not retain context from session to session. The LLM will access organisational knowledge while keeping it confidential. This is also the way to eliminate 'hallucinations', which emerge from creative output and may sound plausible but are factually incorrect. Such content may also not be aligned with the brand or the messaging that marketers are attempting to send.⁴

LLM tools have a parameter called 'temperature' that allows for increasing creativity — including completely fabricated answers. By turning the temperature down to zero and specifying that answers should only come from ingested knowledge, as well as instructing the system to respond with 'I don't know', if the answer is not in the data source, hallucinations can be eliminated. This is called 'retrieval-augmented generation' (RAG).⁵

What follows is a simplified view of how content is processed, ingested and queried by LLMS. First, LLMS require that content be broken up into chunks. Those chunks can vary in size, depending on the LLM and computational requirements (larger chunks require more processing power). However, LLMS use a fixed length of content called a context window. If a piece of content is larger than the context window, it needs to be broken into pieces with overlapping frames — meaning there is overlap between one chunk of content and another. The expectation is that the overlap will provide context for the content.

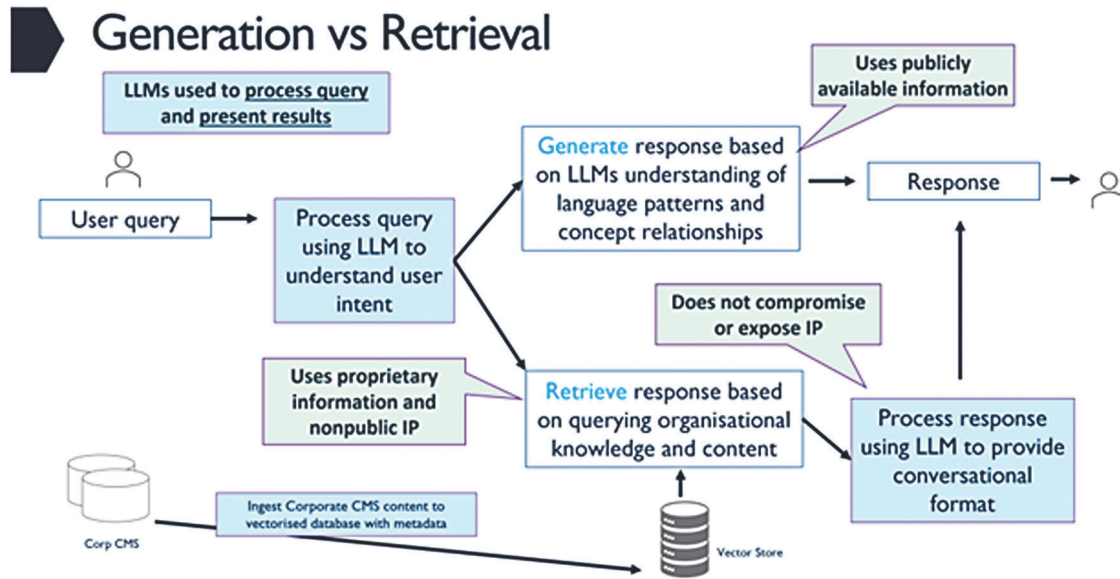


Figure 2: Comparison of generation of answers from an LLM versus retrieval from a knowledge source
Source: Earley Information Science, Inc., 2023

Another way to do this is to intelligently componentise content into more meaningful chunks, such as answers to specific questions or sections of a support manual. Each of these pieces is tagged with metadata that provides context for each component and enables retrieval of a specific answer from a product manual, a policy and procedure knowledge base, or a troubleshooting guide. The metadata improves the performance of the LLM and its ability to answer questions from corporate content sources. The metadata is ingested into the vector store with the content and provides additional signals for retrieval. Figure 3 represents this approach.

LLMs with RAG are on an information continuum that compares other mechanisms of retrieval to LLMs. Figure 4 shows how virtual assistants and chatbots are analogous to search. The user provides a signal in the form of a query, and the system returns a result (a list of documents). Virtual assistants are also retrieval mechanisms meant to define content and functionality more precisely for a specific set of use cases. But any retrieval mechanism shares certain characteristics, as shown in Figure 4.

- Knowledge source: This can be anything from a file share to a well-structured and curated knowledge base. Basic search does not need metadata. In fact, it derives metadata that becomes the search index.
- Search interaction: For a search, the signal is a keyword or phrase, for virtual assistants the interaction is natural language.
- Information architecture: Increasing levels of functionality require more sophisticated information architecture (IA) with ontologies defined. That includes all the taxonomies in the organisation and the relationships between them. Imagine three taxonomies: locations, equipment and maintenance tasks. Equipment at location is one ontological relationship, tasks for maintaining equipment is another.
- User experience: For search, the result is a list of documents perhaps with refiners to filter results. For more precisely designed access such as a role-based knowledge portal, information is carefully curated and presented in context to a task. One portal for Applied Materials saved the company US\$50m per year by reducing the amount of time spent searching and

Componentised Content with Metadata for LLMs

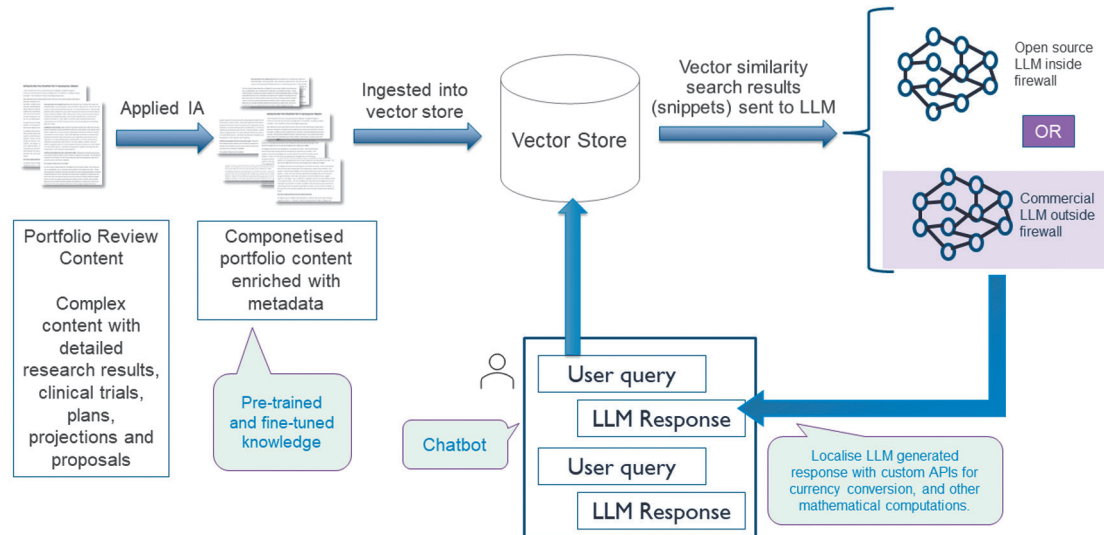


Figure 3: Componentisation and ingestion of content into a vector store enriched by metadata
Source: Earley Information Science, Inc., 2023

Information Retrieval Continuum

	Increasing functionality			
	BASIC SEARCH ENGINE	KNOWLEDGE PORTAL	VIRTUAL AGENT	RETRIEVAL AUGMENTED GENERATION
KNOWLEDGE BASE	Any text Multiple sources	Multiple sources, separate taxonomies and schemas	Domain specific ontologies Highly curated sources	Knowledge ingested into vector space
SEARCH INTERACTION	Keyword or full text query	Full text query or Faceted exploration	Query, explore facets Offers related info	Similarity/semantic search (vector search) Augmented with metadata
INFORMATION ARCHITECTURE	None necessary, but Improves with metadata	Taxonomies, clustering, classification	Ontologies, clustering, classification, NLP	Ontology, knowledge graph, metadata models, NLP
USER EXPERIENCE	Search box, documents list	Role-Based	Conversational	Conversational, retains context, personalised
ENABLING TECHNOLOGY	Search	Search, classification, databases	NLP, search, classification Process engines	LLM, vector store, NLP, semantic search, classification

Figure 4: The information retrieval continuum
Source: Earley Information Science, Inc., 2023

retrieving information for their field service organisation. A virtual assistant or LLM provides a natural language or conversational experience.

- Enabling technology: Classification is part of every type of retrieval tool. Basic search engines cluster content and build indexes, while a knowledge portal might have integrated workflows. LLMs use vector databases and semantic search.

IF THE LLM IS NOT BEING USED FOR THE ANSWER, THEN WHAT ROLE IS IT PLAYING?

Instead of using it to answer the query directly, LLM is used instead to ‘process’ a query and use that processed query to retrieve information from a knowledge source or database. The results are then also processed to make them sound more conversational. The preprocess part can be considered a ‘thesaurus on steroids’ by making variations in requests conceptually the same. Just as chat utterances must be processed so that questions people ask or comments they make that vary in phrasing but mean the same thing can be converted into a standard intent that the system can respond to, LLMs need to do this at the conceptual level. What is the user trying to accomplish? How is that question the same as other types of questions that other users are asking?

The following is a simplified version of the process, but it explains the main concepts. The system represents a query in mathematical terms called a vector. Vectors are multi-dimensional, and the characteristics of a document or piece of content are represented by those dimensions. A body of content can contain thousands or tens of thousands of characteristics. The LLM compares that mathematical representation to the mathematical representations of knowledge ingested from the knowledge base into a vector database. The closest vector is then returned as an answer. The

LLM processes the answer using its knowledge of language to make the answer sound more conversational.

FINE TUNING AN LLM

LLMs are trained for generalised tasks and understand a broad range of concepts and language relationships. In many cases, a more specialised language model is needed to understand technical or industry-specific terminology. Several of these specialised models have been created, and many are open source. Fine tuning a model can range from building a model from scratch for a particular domain (such as life sciences or financial services) to ingesting training content that is specific to the domain and/or the tasks that require enablement.⁶ For example, an LLM can ingest support chat dialogues and use the content to improve performance for those support tasks. Even when an industry-specific model is used, some terms and processes are unique to the organisation.

In addition, organisations can use relationships among terms, concepts, processes, problems, solutions, tasks, customer types, market segments and more that comprise the unique knowledge to differentiate their products in the marketplace. That knowledge can be captured and represented by a corporate ontology, which is the knowledge scaffolding of the organisation. The ontology consists of all of the different taxonomies of the enterprise and the relationships between those taxonomies. For example, in an insurance company, there could be a list of ‘risks’ and a list of ‘regions’ — two independent taxonomies. When ‘risks’ are mapped to ‘regions’, the relationship is ‘risks in region’ and becomes part of the subject–predicate–object structure of an ontology.

Figure 5 illustrates the types of generalised and specialised language models as well as the organisation specific ontology that forms the basis for differentiation in the marketplace. The ontology captures how

Generalised, specialised and corporate language

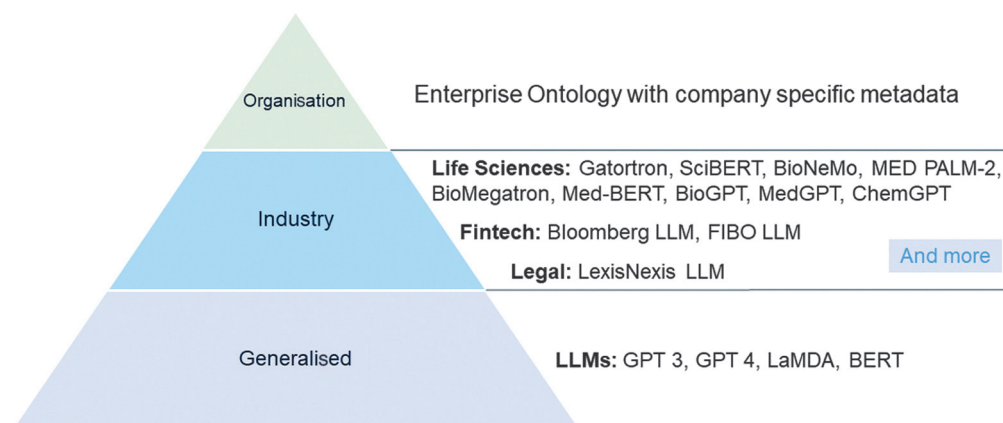


Figure 5: Generalised, industry and organisation specific models and terminology
Source: Earley Information Science, Inc., 2023

the organisation understands the marketplace, competitors, customer needs, solutions, customer types and more.

OPTIONS AND MECHANISMS FOR FINE TUNING

It can be costly to train a model from scratch. Commercial LLMs can also become costly to run. Using an open source LLM behind the corporate firewall can be a viable solution for the enterprise. Figure 6 outlines the trade-offs between commercial and open source LLMs.

Mechanisms for open source and commercial LLMs need to be considered over the short term and long term. A variety of fine-tuning options for open source and commercial LLMs are available, as illustrated in Figure 7.

The first option retains the values and parameters of the commercial LLM, but fine-tunes those based on retraining using corporate knowledge sources and content. However, in this approach, corporate information becomes part of the LLM and is therefore no longer confidential. The second uses retrieval augmented generation

that references the corporate content but does not ingest it into the public model. This safeguards confidential information. In either case, a knowledge/information architecture further improves performance. The way to test this is to have a core set of use cases (queries) with known correct responses and test them against various configurations and models.

The choices for fine tuning can become more varied and nuanced as the various combinations of approaches are considered. Figure 8 outlines the advantages and disadvantages for various options. For each of the options, adding a layer of information architecture improves performance by adding metadata which becomes additional clues for the LLM when responding to queries.

THE VALUE OF METADATA

Many of the vendors offering LLM-based solutions are now recognising the need for knowledge as an essential component; however, some are missing an important detail, which is how to structure the data. The author recently spoke with a generative

Financial and Security Ramifications



Creating an LLM from scratch is cost prohibitive



Fine tuning an LLM can be cost effective

Using commercial LLMs

- Priced per 'token consumed' can be costly
- May violate corporate data policy for information that cannot leave premises

Open source LLMs

- Costs include set up and ongoing management
- Can be set up behind firewall protecting IP and PPII

Figure 6: Financial and security ramifications of commercial and open source LLMs
Source: Earley Information Science, Inc., 2023

To Train or Not to Train



Training costs are drastically lower than even a couple of years ago.

Option 1

Retain previous parameters and weights and fine tune to corporate content to create newer weights

Option 2

Retrieval Augmented Generation (RAG)
Not retraining or changing the model

In either case, enriching with a knowledge architecture adds tremendous value

- Build use cases and standard questions and answers as baseline
- Measure against standardised responses with knowledge/information architecture

Figure 7: Fine-tuning on corporate content versus retrieving corporate content
Source: Earley Information Science, Inc., 2023

AI vendor and asked him about the role of taxonomies, metadata and knowledge graphs. His response was, 'You don't need any of that'. When pressed about how the data was prepared, he admitted, 'Well, we have to do some data labelling'. That said it

all: labelling data is in fact the same thing as applying metadata.

Metadata is more important than many people recognise. In a recent research project, an LLM was found to be able to answer questions 53 per cent of the time

To Train or Not to Train

Approach	Option	Advantages	Disadvantages
Commercial LLM	o Public, no fine tuning	✓ Straightforward	• Possible Exposure of IP • Lack of competitive advantage • Cost
	o Public, with fine tuning	✓ Greater accuracy, precision and recall	• Possible Exposure of IP • Cost
	o RAG	✓ Use of corporate content for ground truth	• Cost
Open source LLM	o Pvt, no fine tuning	✓ Lower LLM cost ✓ Under corporate control	• Higher set up and maintenance costs • Risk of hallucinations
	o Pvt w/RAG and no IA	✓ Lower LLM cost ✓ Under corporate control ✓ Reduce hallucinations	• Higher set up and maintenance costs • Need source of ground truth
	o Pvt w/RAG and IA	✓ Lower LLM cost ✓ Under corporate control ✓ Better recall and precision	• Higher set up and maintenance costs • Need source of ground truth
Open source LLM w/(re)trained model	o Pvt w/RAG, (re)trained model and no IA	✓ Improved accuracy, recall and precision	• More costly and complex
	o Pvt w/RAG, (re)trained model and IA	✓ Best accuracy, recall and precision	• Most costly and complex with ongoing management

Figure 8: Options for fine-tuning commercial and open source LLMs
Source: Earley Information Science, Inc., 2023

without metadata, but 83 per cent of the time with metadata. That is a vast improvement in performance. Figure 9 summarises the importance of information architecture to successful LLM performance. Figure 10 summarises the benefits of augmenting content with metadata for LLMs.

ORGANISING AND CURATING CONTENT

Many believe that LLMs eliminate the need to curate content or apply a defined data and information architecture. That is a misconception that organisations need to address with decision makers. The LLM ‘fix’ seems like a magic bullet, but due to the inherent structures and mechanisms of LLMs, this is not true when accuracy and confidence in answers are necessary for regulatory purposes and to provide a high-quality customer experience. Content is critical, and the old saying of ‘garbage in,

garbage out’ still applies. LLMs need content in the correct context and that context is provided by the additional signals of metadata. Figure 11 summarises this principle.

Results of proof of concept (PoC) using a knowledge architecture

A recent experiment by Earley Information Science found that LLM performance was significantly improved with the approaches being described (RAG with content enriched with metadata). The information was highly sensitive portfolio review content that helped executives make decisions about how to allocate corporate resources for various initiatives. Information included potential acquisitions and the results of various preclinical and clinical trials. The information could not be compromised, and hallucinations were not permissible. Figure 12 shows the situation and sensitivities for the LLM PoC.

The Importance of Information Architecture (IA) in the Age of AI

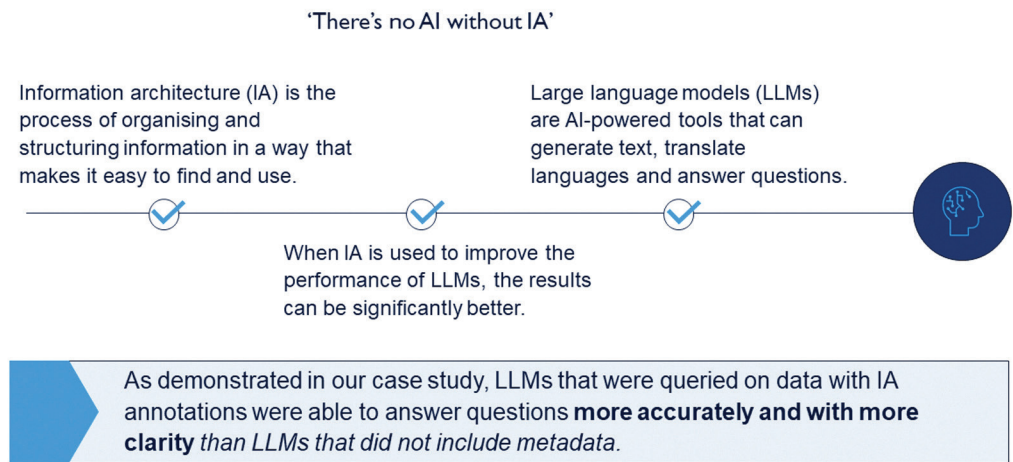


Figure 9: LLMs performance improves with metadata
Source: Earley Information Science, Inc., 2023

How IA Improves the Performance of LLMs

Three ways that IA can make LLMs more effective:

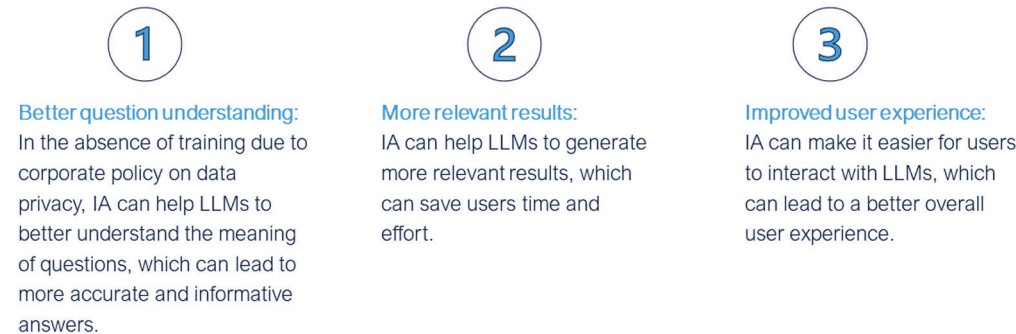


Figure 10: How information architecture (IA) improves LLM performance
Source: Earley Information Science, Inc., 2023

The content included information-dense portfolio narratives and a commercial model was used to process queries and results, but content was not ingested into the LLM. Figure 13 illustrates the PoC test parameters.

The results clearly demonstrated the value of a knowledge/information architecture on LLM performance. As the source system

required that metadata be applied to content, ingesting that metadata into a vector database was not onerous. Many believe that metadata application is onerous and too costly. However, some approaches significantly reduce the level of effort needed. The necessary element is the reference architecture which provides value to multiple tools and technologies in the

Getting your content organised

- ✓ Generalised language models are about understanding the world but will not use your IP for competitive differentiation
- ✓ Content can be ingested into a database that generative AI can use as a reference source
- ✓ If the content is not there, the system cannot provide an accurate answer.
- ✓ This means that content databases are critical to using tools like ChatGPT
- ✓ Metadata (tags) applied to content become additional signals to the LLM
- ✓ Getting an answer to a support question requires context provided by metadata:
 - Product name
 - Model
 - Version
 - Etc. ...

This information (content model) needs to be ingested into a generative AI application as part of the 'embeddings'

Figure 11: Why organising content is important for LLM powered applications
Source: Earley Information Science, Inc., 2023

Case Study: Content Characteristics, Requirements and Purpose

- Characteristic: Portfolio review contains highly sensitive information (acquisitions, drugs in development, clinical trial results, investments, timelines, etc.)
- Characteristic: Pharma is highly regulated industry where hallucinations are not acceptable
- Requirement: Content and data must not be exposed in full to public LLMs
- Requirement: Any shared snippets must not be retained beyond that user session
- **Case study purpose: Determine the impact of IA on LLM's responses**



Figure 12: Requirements of LLM proof of concept
Source: Earley Information Science, Inc., 2023

corporate ecosystem. Figure 14 summarises the results of the PoC with and without metadata.

This data strongly supports the use of a knowledge/information architecture for LLM powered applications. To identify whether this approach is appropriate,

several issues should be assessed in the organisation:

1. Does the organisation educate executives and stakeholders about AI technologies, including capabilities and limitations? It is important to have people on the same

Case Study: Test Parameters

- IA for scoped content – developed content models with metadata schema and applied them to componentised content
- Componentised content - 40+ pages of complex data componentised into logical sections
- Vector store - Local vector store to maintain privacy and protect IP
- LLM Model – Used both open source and commercial models. Results presented here are from the commercial model with 182 billion parameters
- Well-defined use cases – we chose 60 questions typically asked about a portfolio

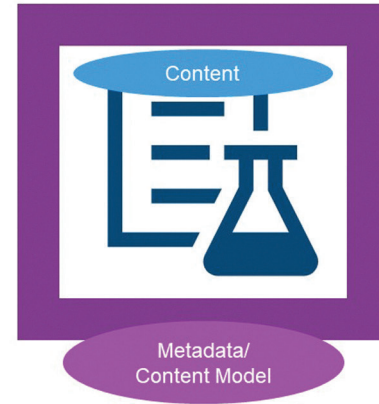


Figure 13: Proof of concept test parameters
Source: Earley Information Science, Inc., 2023

Case Study: Results of Knowledge Architecture on LLM

- IA content models have a significant effect on LLM's responses
- LLM chose to select answers from metadata in content models 79% of the time in this experiment if the answer was in both metadata and content
- IA designed for well-defined business use cases can answer a majority of user's questions.
- LLM's responses were more complete and accurate with IA. In this case study about a third of the questions could not be answered by the LLM without IA.

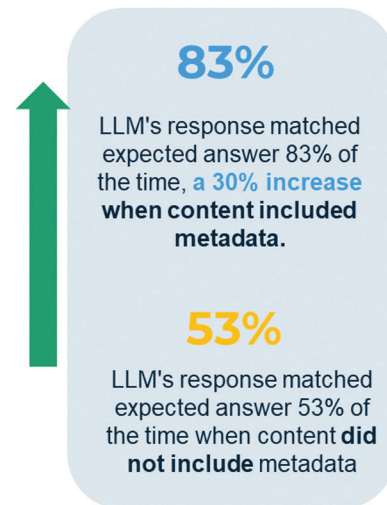


Figure 14: Test results
Source: Earley Information Science, Inc., 2023

- page and level set about what is possible as well as what is practical.
- Has the organisation evaluated the potential business value and target use cases for generative AI?
- What will be measured to determine whether success has been achieved?
- Is the organisation ready for generative AI? The following parameters should be considered:

- business alignment and process clarity;
- knowledge and data readiness and technology infrastructure;
- ongoing governance, decision making and success measures.

Understanding the target use cases and setting and managing expectations will ensure that the correct problems are addressed and that stakeholders agree on success.

Additional guidelines for successfully testing and deploying a ChatGPT type of application are as follows:

1. Define specific use cases: Start with well-defined, narrow use cases. The clearer and more precise, the easier it is to evaluate success. For instance, a vague use case like 'Employ LLM for customer support' does not help. Instead, consider specific tasks such as 'Utilise LLM to guide modem installation referencing the manual' or 'Deploy LLM to identify project milestones using a given project document'.
2. Ensure necessary content availability: For the LLM to function effectively, the relevant content must be available. For instance, if the purpose of the LLM is to guide a technician in modem installations, the detailed installation manual is essential. Similarly, to identify project milestones, the LLM needs access to comprehensive project documentation.
3. Limit creative responses: Adjust the LLM settings to prevent undesired outputs. For instance, set the temperature to zero and direct the model to rely solely on the provided database. If unsure, it should respond with 'I don't know' to avoid generating potentially incorrect information.
4. Maintain a benchmark use case library: Consistently evaluate the LLM's performance using a collection of benchmark use cases. This ensures consistent performance tracking and highlights areas of improvement.
5. Track and address knowledge gaps: Monitor instances where the LLM responds with 'I don't know' to identify content deficiencies. Whenever new data or content sources are introduced, retest these scenarios to ensure gaps are addressed.
6. Enhance data with metadata: Enriching content with metadata boosts LLM efficiency. By tagging content with details like department, topic, content type and other relevant information, the LLM can more accurately respond to queries. This is particularly crucial in scenarios like conversational commerce. Moreover, metadata aids in streamlining other non-AI applications, ensuring smoother technology integrations.
7. Integrate across platforms: While chat is a powerful interface for tasks like sales, consider integrating LLM tools into other systems through APIs. Envisage your knowledge base driving marketing initiatives, e-mails, self-service portals, field services or even being incorporated directly into products.
8. Prioritise user trust: Trust is earned when users comprehend the system's workings. Ensure transparency in how the LLM retrieves and presents information. Demonstrating the logic behind the generated answers will foster trust among stakeholders, from executives to end-users.
9. Establish strong content management and governance: Successful LLM deployment demands robust content operations. Institutions need structured mechanisms for resource allocation, performance assessment and necessary adjustments. While governance might seem less flashy compared to advanced AI functionalities, it is fundamental for a successful deployment.

CONCLUSION

LLMs are revolutionising many aspects of creative work, but they can also have a significant impact on knowledge work. The

organisations that focus on getting their knowledge, data and content house in order will be ahead of the game and be able to capture true competitive advantage in this-fast moving space.

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